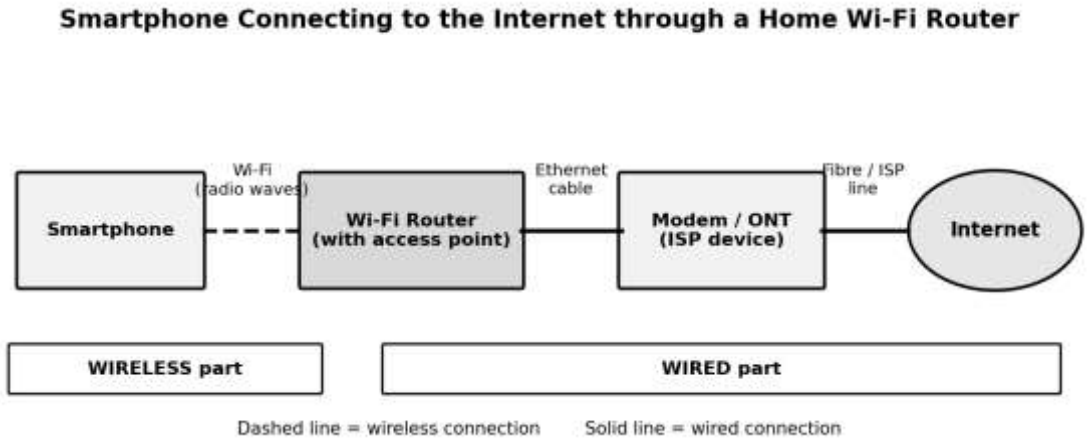
Wireless Networking

Q1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Standard	Wi-Fi name	Year	Frequency band	Max speed (theoretical)	Key difference
802.11n	Wi-Fi 4	2009	2.4 GHz and 5 GHz (dual band)	Up to 600 Mbps	First standard to use MIMO (many antennas) and to support both bands, so it was much faster than older b/g.
802.11ac	Wi-Fi 5	2013	5 GHz only	Up to about 3.5 Gbps	Works only on 5 GHz, but is much faster; it adds wider channels (up to 160 MHz) and MU-MIMO for several devices.
802.11ax	Wi-Fi 6	2019	2.4 GHz and 5 GHz (Wi-Fi 6E adds 6 GHz)	Up to 9.6 Gbps	Works better in crowded places (OFDMA), so many phones and smart devices share the network with less delay, and it saves device battery.

Q2 Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



Part	Type of link	Description
Smartphone to Wi-Fi router	Wireless	The phone sends and receives data as radio waves (2.4 GHz or 5 GHz) to the access point built into the router.
Wi-Fi router to modem / ONT	Wired	An Ethernet cable connects the router's WAN port to the ISP's device (in many homes the router and modem are one box).
Modem / ONT to the internet	Wired	A fibre or telephone line of the ISP (such as Jio or Airtel) carries the data to the internet.

Only the first hop (phone to router) is wireless. All the remaining parts of the path are wired.

Q3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why. Consider both signal strength and security when choosing.

How to scan for networks

- Windows laptop: click the Wi-Fi icon in the taskbar to see available networks. For more details, open Command Prompt and run netsh wlan show networks mode=bssid, which shows the SSID, authentication (security) and signal in percentage.
- Android phone: open Settings, Wi-Fi, and tap a network to see its signal strength and security type.
- iPhone: Settings, Wi-Fi shows the signal bars and the lock icon; the exact security type can be seen after tapping the information icon (i) of a network.

Result of my scan (example entries, replace them with the networks you see)

No.	SSID (network name)	Signal strength	Security type	Remarks
1	Home_WiFi_5G	-45 dBm (excellent, 4 bars)	WPA3 / WPA2 personal	My own router, strong signal and encrypted
2	College_Campus_WiFi	-62 dBm (good, 3 bars)	WPA2 Enterprise	Needs a student login; secure
3	Cafe_Free_WiFi	-70 dBm (fair, 2 bars)	Open (no security)	Free, but anyone can see the traffic

Signal strength guide

Signal (dBm)	Quality	Meaning
-30 to -50	Excellent	Very fast and stable; good for video calls and streaming
-50 to -67	Good	Reliable for browsing and videos
-67 to -80	Fair to weak	Slow speeds and disconnections are possible
Below -80	Very poor	Not usable

Which network I would choose and why

I would choose Home_WiFi_5G. It has the strongest signal, so the speed is higher and the connection is stable. It also uses WPA3/WPA2 encryption, so my data (for example UPI payments and passwords) is protected. College_Campus_WiFi is also a safe choice because of its login, but its signal is weaker. I would avoid Cafe_Free_WiFi, because an open network has no encryption and other users on it could capture my data, even though it is free. If I must use such a network, I should avoid banking and use a VPN.

Screenshot of my scan:

Paste a screenshot of your Wi-Fi scan here

Q4 Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

What is SSID hiding?

The SSID (Service Set Identifier) is the name of a Wi-Fi network, such as Home_WiFi_5G. Normally the router keeps broadcasting this name in small signals called beacon frames, so that it appears in the list of available networks on phones and laptops. SSID hiding (disabling the SSID broadcast) is a router setting that stops the router from announcing the name. The network no longer appears in the list, and a person must type the exact SSID and password manually to connect.

Advantage: It keeps the network out of sight of neighbours and casual users, so fewer people try to connect or guess the password, and the list of networks in a crowded place such as a hostel or apartment looks cleaner.

Disadvantage: It does not give real security. The SSID can still be found easily with free tools such as Wireshark, because the name is sent when a device connects, and devices keep searching for the hidden network, which can reveal its name. It is also inconvenient, because every new device needs the name typed in correctly.

Conclusion: Hiding the SSID is only a small extra step and not a replacement for security. The network should always be protected with a strong password and WPA2 or WPA3 encryption.